

Rebuttal to Reviewer 2kXw

We sincerely thank you for the positive assessment of the paper, especially for recognizing the clarity of exposition, the detailed mathematical development, and the value of using Gumbel-Softmax and Binary-Concrete relaxations to bridge discrete program synthesis with gradient-based variational inference. We have revised the original submission to directly address your concerns regarding experimental scope, the choice of maximum tree depth, and the scalability of memory footprint as well as gradient variance of relaxations with tree depth.

(1) Results of VaSST and competing methods for learning the newly added Feynman equation $f = \beta(1 + \cos \theta)$. Regarding the evaluation on the Feynman Symbolic regression Database (FSReD), we agree that a full evaluation over all 100+ equations would be a valuable large-scale benchmark. Our goal in the original submission, however, was to demonstrate the performance of VaSST as a probabilistic symbolic regression method on a carefully chosen and scientifically meaningful suite of Feynman equations spanning different levels of structural complexity. In particular, the originally reported equations-CL, CPE, FCE, and FTC-were selected because they collectively represent a robust test suite containing multiplicative interactions, rational forms, additive decompositions, and trigonometric structure. Thus, these examples are not arbitrary isolated cases, but rather a diverse set of symbolic regimes designed to evaluate whether VaSST can recover the underlying scientific structure and physical truth hidden in noisy data.

Importantly, VaSST is not restricted to these examples: the implementation directly extends to all equations in the FSReD. Following your suggestion, and to further strengthen the empirical evidence on the revision, we have added a new Feynman equation

$$f = \beta(1 + \cos \theta),$$

corresponding to the angular distribution with forward-backward symmetry. This equation provides an additional test of VaSST’s ability to recover a physically interpretable baseline-plus-trigonometric structure. Across different noise levels, VaSST successfully recovers the underlying symbolic form with RMSE close to the injected noise level, further supporting the robustness of the proposed probabilistic symbolic regression framework.

In addition to the competing symbolic regression methods considered in the original submission, we have also added comparisons with several widely used modern symbolic regression methods, including Operon (Burlacu et al., 2020), PySR (Cranmer, 2023), and DSR (Petersen et al., 2019) (as per your suggestion). The results for VaSST and all competing methods on the newly added equation $f = \beta(1 + \alpha \cos \theta)$ across different noise levels are reported below.

Method	Learned expression	RMSE
VaSST	$f = -0.8994 + 0.00125 \cos(2\alpha) + 0.10297\theta$ $+ \underline{1.02384 \beta(1 + \alpha \cos \theta)}$	0.13497
PySR	$f = \beta \left[0.009903 \alpha (\theta^2 - 24.691) (7.917 - \theta) + 0.29063 \right] (10.354 - \theta)$	0.45075
Operon	$f = 0.214 - 3.021\beta \left[e^{-0.484\theta} + 0.083\alpha + \frac{e^{-0.500\theta}}{0.080\alpha} \right.$ $\left. - \sin(\sin(0.757\theta)) \{ \sin(0.757\theta) - 0.404\alpha \} \right]$	0.66695
DSR	$f = \beta(1 + \alpha \cos \theta)$	1.848×10^{-15}

Table 1: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ in the noiseless setting from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
BMS	$\hat{f} = \beta \{ 1 + \alpha \cos \theta \}$	1.76×10^{-15}
BSR	$f = -0.229931 + 6.94561 e^{\cos \theta} + 0.225102 \beta^2 - 0.00193193 e^{\theta}$	6.92292
QLattice	$f = 324.462 \left[-0.0214663\alpha + \exp\{-1.90296(0.0865876\alpha - 1)^2 - 4.48049(0.203294\theta - 1)^2\} + 0.0167656 \right]$ $\times \exp\{-2.14832(1 - 0.140906\beta)^2 - 10.6876(0.111068\theta - 1)^2\} + 0.0363653$	0.48031
gplearn	$f = \left[\theta\alpha + \frac{-0.069}{(\beta - 0.593)(0.531 + \beta - \theta)\beta(\beta - \theta) + \beta\theta} \right.$ $+ \frac{\theta\alpha + \frac{-0.069}{(\beta - 0.593)(\beta - \theta)} + \frac{\theta}{\theta(\alpha / -0.731)} + (\beta - 0.593)(0.531 + \beta - \theta) + \alpha - \frac{\alpha}{-0.731}}{\theta} + 0.445\theta + \alpha \left. \right]$ $\times \left[\frac{\frac{\beta}{\alpha} + \theta + \beta + (\beta - 0.593)(\beta - \theta)}{(0.531 + \beta)\beta} + 0.445\theta \right] (-0.010\theta)$ $+ 0.530\alpha \left[\frac{\theta}{\alpha} + \theta + \beta + (\beta - 0.593)(0.531 + \beta - \theta) \right] + 2\alpha$	6.31679
DEAP	$f = \frac{\alpha^2}{\theta + \left(\frac{\beta}{\theta} - \beta \right)} = \frac{\alpha^2}{\theta + \frac{\beta}{\theta} - \beta}$	13.54156

Table 2: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ in the noiseless setting from BMS, BSR, QLattice, gplearn, and DEAP along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$f = -0.92928 + 0.000575 \cos(2\alpha) + 0.10661\theta$ $+ \frac{1.02407 \beta(1 + \alpha \cos \theta)}{1}$	0.16513
PySR	$f = (-2.48391\theta + 20.96369 + \alpha)$ $\times \beta(\theta - 4.86540)(-0.24211\theta + 1.95075)$	1.49573
Operon	$f = 0.377 - 83.187 \left[\frac{\cos(-0.981\theta)(-0.368\beta)}{\sqrt{1 + (-1.085\theta)^2}} \right.$ $\times \left. \frac{1}{\sqrt{1 + \left(\frac{\cos(\cos(\cos(-0.999\theta)))}{\sqrt{1 + \exp\{\cos(-0.999\theta) \cos(0.340\alpha)\}^2}} \right)^2}} \right] (0.318\alpha)$	0.51157
DSR	$f = \beta(1 + \alpha \cos \theta)$	0.09908

Table 3: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under noise $\sigma = 0.1$ from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
BMS	$f = \beta(1 + \alpha \cos \theta)$	0.10022
BSR	$f = -0.232683 + 6.94597e^{\cos \theta} + 0.225131\beta^2 - 0.00193157e^{\theta}$	6.92420
QLattice	$f = -0.619856\alpha + 19.747(3.03165 - 0.377702\theta)(1.28896\beta + 0.0343982)$ $\times \exp \left\{ -12.6002(1 - 0.129159\theta)^2 - 2.02944(0.0893216\alpha - 1)^2 \right\} + 3.45018$	0.53195
gplearn	$f = [\theta\alpha + \{(-0.097 + \beta)(\alpha - \beta) - (\theta + \beta) - \theta\}] \left[\left(\frac{0.649}{\theta} + 0.445\theta \right) (-0.010\theta) \right]$ $+ \beta(\beta - 0.593)(\beta - \theta) \left[(-0.097 + \beta) \left\{ \frac{\alpha - \beta}{\theta} - \frac{-0.069}{(0.531 + \beta) \left[\frac{-0.069}{\beta + \frac{\beta\{0.833 + \beta - \frac{\theta - (\theta - \beta)}{\beta\theta}\}}}{\beta\theta}} \right] + 0.247 \right\} + \theta \right]$ $+ 2\alpha$	6.36929
DEAP	$f = \frac{\alpha^2}{\theta + \left(\frac{\beta}{\theta} - \beta \right)} = \frac{\alpha^2}{\theta + \frac{\beta}{\theta} - \beta}$	13.54231

Table 4: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under noise level $\sigma = 0.1$ from BMS, BSR, QLattice, gplearn, and DEAP along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$f = -0.95916 - 0.000097 \cos(2\alpha) + 0.11026\theta$ $+ \underline{1.02430 \beta(1 + \alpha \cos \theta)}$	0.23748
PySR	$f = \left[-0.02377 \left\{ \theta + -0.70132(\theta - 9.13492)^2 \right\}^2 + 1.22422 \right] \alpha \beta$	1.32648
Operon	$f = 3.429 + 0.352 \left[\left\{ \frac{\cos(0.976\theta)}{\sqrt{1 + (0.653\theta)^2}} \right\} (5.549\beta)(2.578\alpha) \right.$ $\left. + \left\{ \frac{\cos(-0.781\theta)}{\exp(\cos(1.718\theta))} \right\} (-0.437\alpha) - 1.536\alpha \right]$	0.46428
DSR	$f = \beta(1 + \alpha \cos \theta)$	0.1981

Table 5: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under noise $\sigma = 0.2$ from VaSST, PySR, Operon, and DSR along with their RMSEs.

Method	Learned expression	RMSE
BMS	$f = \beta(1 + \alpha \cos \theta)$	0.20045
BSR	$f = -0.235435 + 6.94633e^{\cos \theta} + 0.225159\beta^2 - 0.00193121e^\theta$	6.92692
QLattice	$f = -68.9593(0.00212903 - 0.280842\beta)$ $\times \left[-0.0561226\alpha + \exp\{-2.07305(1 - 0.0903278\alpha)^2 \right.$ $\left. - 3.37702(-0.109637\theta - 1)^2(0.157901\theta - 1)^2\} + 0.0526521 \right] + 0.0674459$	0.50443
gplearn	$f = \{\theta\alpha + (\theta - \beta)\} \left[\left(\frac{0.649}{\theta} + 0.445\theta \right) (-0.010\theta) \right]$ $+ \beta(\beta - 0.593)(\beta - \theta) \left[(-0.097 + \beta)A + \theta + B \right] + 2\alpha$	6.42362
DEAP	$f = \frac{\alpha^2}{\theta + \left(\frac{\beta}{\theta} - \beta \right)}$	13.54380

Table 6: Learned symbolic expressions for Feynman equation $f = \beta(1 + \alpha \cos \theta)$ under noise level $\sigma = 0.2$ from BMS, BSR, QLattice, gplearn, and DEAP, along with their RMSEs. For compactness, the gplearn expression uses the shorthands $A = \frac{\alpha - \beta}{\theta} - \frac{-0.069}{\beta\theta}$, and $B = 0.833 + \beta - [\theta - (\theta - \beta)] + \frac{\theta\beta + \theta/\beta}{\theta}$, where $C = \left[(-0.097 + \beta) \left(\frac{\alpha - \beta}{\theta} - \frac{-0.069}{\frac{-0.097 + \beta}{\frac{0.649}{\theta} + 0.445\theta}} \right) + \theta + \beta \right] (\beta - 0.593)(\beta - \theta)/\beta\theta + \frac{-0.069}{(\beta - \theta)(\beta + 0.247) + \frac{-0.097 + \beta}{D}}$ with $D = \left[\left(\frac{-0.069}{\beta + \frac{\beta\{0.833 + \beta - [\theta - (\theta - \beta)]\}}{\beta\theta}} \right) + 0.247 \right] (\beta + 0.247)(\alpha - \beta)$.

(2) On Depth D and Ablation Study. We thank you for raising this important point. In VaSST, the number of trees K and maximum depth D are user-specified capacity parameters controlling the richness of the symbolic ensemble. The choice $(K, D) = (3, 3)$ in the original

experiments was not meant to be a methodological restriction. Rather, it was chosen because the Feynman equations considered in the original submission could already be represented accurately with a small number of moderately shallow trees. Since VaSST uses the additive representation

$$y_i = \beta_0 + \sum_{k=1}^K \beta_k T_k(x_i) + \varepsilon_i, \quad i = 1, 2, \dots, n,$$

a complex scientific expression need not be represented by one deep tree; it can be decomposed into several interpretable symbolic components. Thus, $D = 3$ provided a favorable accuracy-interpretability-runtime trade-off for the original experiments considered.

To make this explicit, we have added an ablation study over $K \in \{2, 3, 4, 5\}$ and $D \in \{2, 3, 4, 5\}$ for all the Feynman equations, where these new results will feature in the Appendix, along with a discussion of VaSST’s sensitivity with respect to K and D . A highlight of the ablation study for two representative Feynman equations

$$P = \frac{\kappa A(T_2 - T_1)}{d}, \quad \Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right),$$

is as follows. Gaussian noise with standard deviation 0.1 has been added to the data. See Tables 7–10. The key findings are:

- **Performance stabilizes with model complexity.** For both equations, RMSE improves significantly when moving from $K = 2$ to $K \geq 3$, and stabilizes for $K \geq 4, D \geq 3$. This indicates that VaSST does not require large model sizes to achieve optimal predictive accuracy.
- **Low-capacity regimes can limit symbolic recovery.** As expected, insufficient model capacity leads to occasional failures in recovering the true expression. For

$$P = \frac{\kappa A(T_2 - T_1)}{d},$$

failure occurs at $(K = 2, D = 5)$, and for

$$\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right),$$

failures occur at $(K = 2, D \in \{2, 4, 5\})$ and $(K = 3, D = 5)$. This highlights that both sufficient K and appropriate D are necessary for capturing compositional structure.

- **Runtime scales predictably with K and D .** Runtime increases approximately linearly in K and exponentially with D , reflecting the growth in the number of nodes. Importantly, moderate configurations (e.g., $K = 3, D = 3$) already achieve near-optimal RMSE, offering a favorable accuracy–efficiency trade-off.
- **Principled heuristics for choosing K and D .** Based on the ablation results, we include practical guidelines for selecting K and D . Since VaSST is designed to recover simple and interpretable scientific expressions, our recommendation follows an Occam’s razor principle: choose the smallest model capacity that attains stable RMSE and symbolic recovery. Empirically, $K \in \{3, 4\}$ and $D \in \{3, 4\}$ provide a favorable accuracy–interpretability–runtime trade-off, while larger depths increase runtime exponentially and may lead to unnecessarily complex symbolic expressions without substantial predictive gains.

Table 7: Runtime for $P = \frac{\kappa A(T_2 - T_1)}{d}$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	50.6	97.2	191.3	401.6
3	62.1	112.0	221.8	498.5
4	72.2	125.0	249.8	628.2
5	83.1	146.0	299.4	803.4

Table 8: RMSE for $P = \frac{\kappa A(T_2 - T_1)}{d}$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	0.1019	0.1002	0.1002	0.1070
3	0.1000	0.1011	0.1012	0.1019
4	0.1000	0.0999	0.1000	0.1000
5	0.0999	0.0999	0.0998	0.1007

Table 9: Runtime for $\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	54.7	98.8	192.4	399.6
3	61.5	112.0	221.4	491.3
4	67.9	123.1	251.8	600.9
5	72.3	134.8	286.4	800.5

Table 10: RMSE for $\Delta U = Gm_1m_2 \left(\frac{1}{r_2} - \frac{1}{r_1} \right)$ across number of trees K and depth D

$K \backslash D$	2	3	4	5
2	0.2330	0.1012	0.1959	0.2349
3	0.1002	0.1019	0.1051	0.1918
4	0.1168	0.1002	0.1005	0.1012
5	0.1002	0.1005	0.1004	0.1010

(3) Memory Footprint and Gradient Variance. To directly address your question on how the memory footprint and gradient variance scales as the maximum depth D increases, we have added a diagnostic experiment for the Feynman equation (which will feature in the Appendix and will be discussed in the revision)

$$\Delta U = Gm_1m_2(1/r_2 - 1/r_1),$$

using $K = 3$ trees and depths $D \in \{2, 3, 4, 5\}$. The goal of this experiment is to isolate the computational and stochastic-gradient behavior induced by increasing the depth of the relaxed binary tree skeleton.

Mathematically, a full binary tree of maximum depth D contains $N = 2^{D+1} - 1$ nodes. For each tree $k \in \{1, 2, \dots, K\}$, VaSST maintains relaxed logits for node expansion, operator selection, and feature/terminal selection. If the operator set has cardinality $M = |\mathcal{O}|$ and the input dimension is p , then the total number of relaxed logits scales as $d_{\text{relax}}(K, D) = KN_D(1 + M + p)$. Thus, for fixed K, M, p , the number of relaxed variational parameters grows as $d_{\text{relax}}(K, D) = O(K2^D)$.

This gives the dominant depth-dependent memory footprint for the structural relaxation. In addition, during the Monte Carlo (MC) ELBO evaluation, the soft-tree forward pass stores node-level values and stochastic relaxation samples. With n training samples and S MC samples, the activation-level memory is of order $O(SnKN)$, up to operator-specific constants. Therefore, the total memory footprint grows approximately linearly in the number of trees K , the number of MC samples S , and the sample size n , but exponentially in depth D .

The same scaling also explains the runtime behavior. A single stochastic forward-backward pass requires evaluating all relaxed nodes and backpropagating through the Gumbel-Softmax/Binary-Concrete relaxation. Hence the computational cost per iteration is approximately $O(SnKNC_O)$, where C_O denotes the average cost of evaluating the operator mixture. Since $N = 2^{D+1} - 1$, this cost grows exponentially with D .

For gradient variability, let $\lambda = (\lambda^{(e)}, \lambda^{(\text{op})}, \lambda^{(\text{ft})}) \in \mathbb{R}^{d_{\text{relax}}}$ denote the collection of relaxed structural logits, consisting of expansion logits, operator logits, and feature logits. For one MC draw of the relaxed tree variables, the stochastic gradient of the negative ELBO $g(\lambda; \omega) = \nabla_{\lambda} \{-\hat{\mathcal{L}}(\lambda; \omega)\}$, where ω denotes the Gumbel-Softmax/Binary-Concrete randomness. We estimate the coordinate-wise gradient variance by drawing R independent stochastic gradient $g^{(1)}, \dots, g^{(R)}$ and computing

$$\widehat{\text{Var}}_j(g) = \frac{1}{R-1} \sum_{r=1}^R (g_j^{(r)} - \bar{g}_j)^2, \quad \bar{g}_j = \frac{1}{R} \sum_{r=1}^R g_j^{(r)}.$$

We then report two summaries

$$\widehat{V}_{\text{coord}} = \frac{1}{d_{\text{relax}}} \sum_{j=1}^{d_{\text{relax}}} \widehat{\text{Var}}_j(g),$$

the mean coordinate-wise gradient variance, and

$$\widehat{V}_{\text{total}} = \sum_{j=1}^{d_{\text{relax}}} \widehat{\text{Var}}_j(g),$$

the total gradient variance across all relaxed structural parameters. The first quantity measures the average stochastic fluctuation per relaxed coordinate, whereas the second measures the aggregate stochastic fluctuation over the full relaxed symbolic parameterization.

The diagnostic experiment was run on CPU for the Feynman equation as stated above with an added noise level of $\sigma = 0.2$. The training samples were $n = 2000$, $K = 3$, and depths $D \in \{2, 3, 4, 5\}$. For this experiment, we configure the operator set as $\mathcal{O} = \{\times, /, +\}$. For each depth, we measured the runtime of one stochastic forward-backward pass using $S = 8$ MC samples in the ELBO. For gradient variance, we generated $R = 100$ independent stochastic gradients using one MC sample per gradient estimate. Gradients were collected only with respect to the relaxed structural logits, namely the expansion logits, operator logits, and feature logits. Each diagnostic was repeated 5 times, and the median value was reported. We computed these quantities both before training and after 2000 training iterations, using the experimental settings of VaSST as mentioned in Appendix H.1 of the original submission.

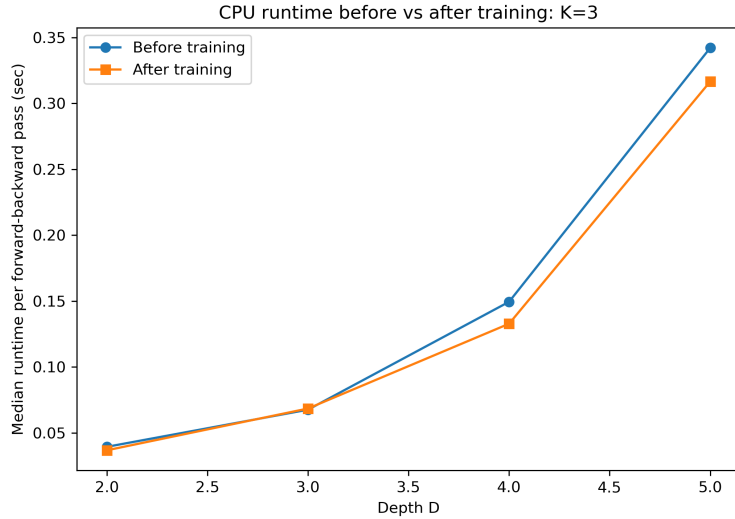


Figure 1: Median (over 5 repetitions) CPU runtime per forward-backward pass against depth D , before and after training for $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$ with $K = 3$ and $\sigma = 0.2$.

The runtime results (Figure 1) confirm the theoretical scaling. The median CPU time per forward-backward pass increases monotonically with depth. This growth is consistent with the expansion of the full tree skeleton from $N = 2^{2+1} - 1 = 7$ to $N = 2^{5+1} - 1 = 63$ nodes per tree. The before-training and after-training runtime curves are nearly identical, indicating that computational cost is governed primarily by the size of the relaxed skeleton rather than by the learned parameter values.

The gradient variance results (Figure 2a) provide a useful clarification. Increasing D does not lead to an explosion of the Gumbel-Softmax gradient variance. Instead, the mean coordinate-wise gradient variance decreases as depth increases. This behavior is consistent with the depth-dependent split prior: deeper nodes have smaller prior probability of being active, so many deep expansion/operator/feature logits receive weaker effective gradients and therefore contribute smaller coordinate-wise fluctuations. In other words, although increasing D adds many relaxed coordinates, many of these coordinates correspond to deeper nodes that are strongly regularized toward inactivity.

The total gradient variance (Figure 2b) also decreases over the tested range of depths, from the order of 6×10^6 at $D = 2$ to approximately 10^6 at $D = 5$, both before and after training. After training, the total gradient variance is slightly larger than before training, but the same decreasing trend with D is preserved. Thus, for this representative equation, increasing depth mainly increases the deterministic computational and memory burden through the exponential growth of the full

skeleton; it does not destabilize optimization through exploding stochastic gradient variance.

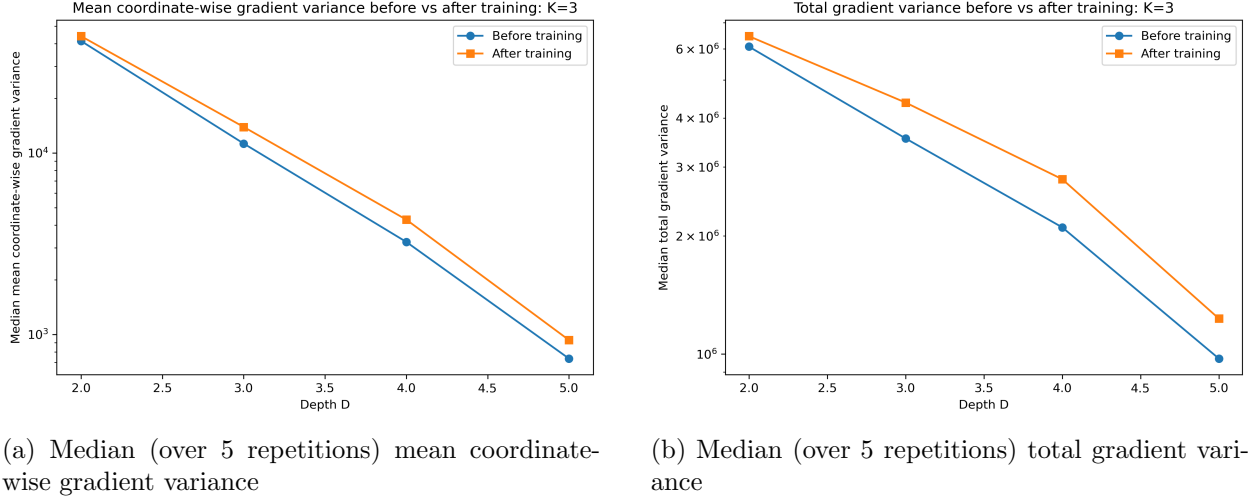


Figure 2: Gradient variance against depth D , before and after training for $\Delta U = Gm_1m_2(1/r_2 - 1/r_1)$ with $K = 3$ and $\sigma = 0.2$.

These diagnostics justify the use of moderate depths in the main experiments. Although larger D expands the symbolic search space and allows more complex expressions, the memory and runtime scale as $O(K2^D)$, and the ablation study shows that $D=3$ already suffices to recover the relevant scientific structures for the equations considered. Therefore, $K = 3, D = 3$ offers a principled accuracy–interpretability–efficiency trade-off, while larger depths can be used when the scientific problem requires more compositional complexity.

(4) Results of VaSST and DSR. We thank you for the valuable suggestion to compare VaSST with DSR (Petersen et al., 2019). Following this suggestion, we have added new experiments using DSR in the revision across all Feynman equations. Here, specifically we compare VaSST and DSR on representative Feynman equations, including the newly added angular-distribution equation discussed above, as well as the equations

$$\Delta U = Gm_1m_2(1/r_2 - 1/r_1), \quad P = \kappa A(T_2 - T_1)/d,$$

across varying levels of noise.

For VaSST, we used $K = 3$ and $D = 3$ with the operator set $\mathcal{O} = \{+, \times, -, /, \exp, \sin, \cos, \log, ^2\}$ along with total observations $n = 2000$. During training, we report the top learned symbolic expression ranked by the in-sample RMSE. Then using this learned model we compute the out-of-sample RMSE on the 10% held-out test set. All remaining hyperparameters were configured as in Appendix H.1 of the original submission. For DSR, we used the `DeepSymbolicRegressor` implementation (<https://github.com/dso-org/deep-symbolic-optimization>), using the same operator set as VaSST. The training budget was set to `n.samples = 200000` candidate expressions with `batch.size = 500`, and the policy length was set to `max.length = 20`.

The results below along with the results for the newly added Feynman equation as presented above show that both VaSST and DSR perform competitively (across all noise levels considered) in recovering the true underlying symbolic expression with RMSE closely tracking the injected noise level. However, observe that VaSST shows a modest degradation in predictive accuracy while learning $f = \beta(1 + \cos \theta)$, but still achieves strong structural recovery by identifying the core component $\beta(1 + \alpha \cos \theta)$.

Method	Learned expression	RMSE
VaSST	$\Delta U = -0.000714 + 0.000914 \frac{G^2 m_1^3 m_2}{r_1^2} - 0.996541 \frac{G m_1 m_2}{r_1} + 1.010681 \frac{G m_1 m_2}{r_2}$	0.003656
DSR	$\Delta U = \frac{G m_1 m_2}{r_2} - \frac{G m_1 m_2}{r_1}$	8.70×10^{-16}

Table 11: Learned symbolic expressions for Feynman equation $\Delta U = G m_1 m_2 (1/r_2 - 1/r_1)$ in the noiseless setting from VaSST and DSR along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$\Delta U = -0.003874 + 0.000609 \frac{G^2 m_1^3 m_2}{r_1^2} - 0.995991 \frac{G m_1 m_2}{r_1} + 1.017123 \frac{G m_1 m_2}{r_2}$	0.100194
DSR	$\Delta U = \frac{G m_1 m_2}{r_2} - \frac{G m_1 m_2}{r_1}$	0.09908

Table 12: Learned symbolic expressions for Feynman equation $\Delta U = G m_1 m_2 (1/r_2 - 1/r_1)$ under noise $\sigma = 0.1$ from VaSST and DSR along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$P = 0.014782 - 0.805371 \left[\frac{A T_1 \kappa}{d} \right] + 0.998520 \left[\frac{A \kappa T_2}{d} \right] - 0.000254 \left[\frac{1}{\sqrt{A T_1 \kappa}} - \frac{\frac{d}{\sqrt{A}} - \frac{T_2}{T_1}}{\sqrt{A T_1 \kappa} - \frac{T_2}{T_1}} \right]$	0.012602
DSR	$P = \frac{A \kappa (T_2 - T_1)}{d}$	1.61×10^{-16}

Table 13: Learned symbolic expressions for Feynman equation $P = \kappa A (T_2 - T_1)/d$ in the noiseless setting from VaSST and DSR along with their RMSEs.

Method	Learned expression	RMSE
VaSST	$P = 0.019505 - 0.842846 \left[\frac{AT_1\kappa}{d} \right] + 1.005597 \left[\frac{A\kappa T_2}{d} \right]$ $- 0.000149 \left[\frac{1}{\sqrt{AT_1\kappa}} - \frac{\frac{d}{\sqrt{A}} - \frac{T_2}{T_1}}{\sqrt{AT_1\kappa} - \frac{T_2}{T_1}} \right]$	0.100325
DSR	$P = \frac{A\kappa(T_2 - T_1)}{d}$	0.09908

Table 14: Learned symbolic expressions for Feynman equation $P = \kappa A(T_2 - T_1)/d$ under noise $\sigma = 0.1$ from VaSST and DSR along with their RMSEs.

(5) Results of VaSST and DSR in learning $P = \kappa A(T_2 - T_1)/d$ under increased noise of $\sigma = 2$. More importantly, we further increased the noise level to $\sigma = 2$ for the equation $P = \kappa A(T_2 - T_1)/d$ to examine robustness in a substantially noisier regime. In this setting, VaSST, using $K = 4, D = 3$, still identifies the physically correct structural components: $T_0 = AT_2\kappa/d$ and $T_1 = AT_1\kappa/d$ with proper estimated regression coefficients corresponding to T_0 and T_1 . In contrast, DSR returns an expression which achieves a similar RMSE but does not recover the correct scientific law. This comparison is particularly important: under high observational noise, purely RMSE-driven symbolic search can identify expressions that fit the data but obscure the underlying scientific mechanism. VaSST, by contrast, continues to recover the physically meaningful symbolic components, illustrating the advantage of its probabilistic, posterior-based formulation and its parsimony-inducing tree prior.

Method	Learned expression	RMSE
VaSST	$P = 0.0153439 + 1.163137 T_0 - 1.151900 T_1 + 0.048648 T_2 + 0.0793742 T_3,$ $T_0 = \frac{AT_2\kappa}{d},$ $T_1 = \frac{AT_1\kappa}{d},$ $T_2 = \frac{d(T_2 - T_1)}{\sqrt{AT_1\kappa}},$ $T_3 = \frac{\frac{d}{\sqrt{A}} - \sqrt{AT_1\kappa}}{\sqrt{AT_1\kappa} \left(\frac{\sqrt{A}}{d} - \frac{T_2}{T_1} \right)}.$	1.99425
DSR	$P = \frac{2(\sqrt{AT_1\kappa}) \left(\frac{T_2}{T_1} \right)^2}{\left(\frac{d}{\sqrt{A}} \right) \left[\sqrt{AT_1\kappa} + 2 \frac{T_2}{T_1} \right]}.$	1.97941

Table 15: Learned symbolic expressions for Feynman equation $P = A\kappa(T_2 - T_1)/d$ under increased noise $\sigma = 2$ from VaSST ($K = 4, D = 3$) and DSR along with their RMSEs.